

CricMetrics Pro: Complete Machine Learning Models Guide

Easy-to-Understand Guide to Every Model Used in This Project (What, When, Why, How)

About this Guide: This document explains all the Machine Learning (ML) models used in the **CricMetrics Pro** cricket player performance system. It is written in simple, clear, and easy-to-understand English so that anyone—students, evaluators, coaches, and team managers—can easily understand how each model works, why it was chosen, and how it performs.

1. Quick Summary of All Models

In this project, we analyze **619 real IPL cricketers** across **17 seasons (2008–2024)** using **260,920 deliveries**. We evaluate **9 Machine Learning models** across 4 core machine learning tasks:

ML Task	Goal / Output	Models Implemented	Top Performer
1. Continuous Regression	Predict continuous Overall Rating (50.0 - 95.0) & Fair Auction Purse (₹ Crores)	- Random Forest Regressor - Polynomial Regression (Deg 2) - Linear Regression (OLS)	Polynomial Regression ($R^2 = 0.9941$, RMSE = 0.6648) & Random Forest ($R^2 = 0.9780$, RMSE = 1.2837)
2. Multi-Class Classification	Categorize player into 3 Talent Tiers: - Elite / Marquee (Tier 1) - Core / Star (Tier 2) - Developing / Squad (Tier 0)	- K-Nearest Neighbors (KNN) - Random Forest Classifier - Logistic Regression - Decision Tree (CART)	K-Nearest Neighbors (KNN) (95.16% Test Acc, 0.9501 Macro F1) & Random Forest (94.35% Test Acc, 94.75% 5-Fold CV)
3. Unsupervised Clustering	Group players into natural playing styles without using predefined human labels	K-Means Clustering ($k=5$)	5 Tactical Archetypes (Anchor, Finisher, Fast Bowler, Spinner, All-Rounder)
4. Dimensionality Reduction	Compress 29 complex statistics into an interpretable 2D tactical map	Principal Component Analysis (PCA)	2D Latent Map (60.79% Total Variance Explained)
5. Data Preprocessing	Clean, encode, and scale features fairly	- StandardScaler (Z-score scaling) - One-Hot Encoding	StandardScaler ($\mu = 0$, $\sigma = 1$)

2. Data Preparation Tools

Before feeding cricket numbers into machine learning models, we must prepare the data so algorithms can compute distances and weights fairly.

2.1 StandardScaler (Z-Score Normalization)

- WHAT is it?

A mathematical tool that rescales all numerical features so that their mean is 0 and standard deviation is 1:

$$z = \frac{x - \mu}{\sigma}$$

- **WHEN is it used?**

Immediately after splitting the dataset into training (80%) and testing (20%). It is fitted **only on training data** and then used to scale test data and real-time user inputs in the web app.

- **WHY do we need it?**

In cricket, `total_runs` can be thousands (e.g. 8,000 runs for Virat Kohli), while `economy_rate` is small (e.g. 7.2 runs per over). If we do not scale them, distance-based models (KNN, K-Means, PCA) will pay 99% of their attention to runs and completely ignore bowling economy. Scaling ensures every skill contributes proportionally.

- **HOW is it implemented?**

Implemented via Scikit-Learn `StandardScaler()` and saved to disk as `models/scaler.joblib`.

2.2 One-Hot Encoding

- **WHAT is it?**

A transformation that converts categorical text (like `'All-Rounder'` or `'Specialist Bowler'`) into orthogonal binary indicator columns (0 or 1).

- **WHEN is it used?**

During the data pipeline transformation in `src/train_cricket_models.py` and `app.py`.

- **WHY do we need it?**

Algorithms cannot read text strings. If we labeled roles as `1 = Batter, 2 = All-Rounder, 3 = Bowler`, algorithms would assume a Bowler is "three times" a Batter. One-hot encoding creates separate switches (`primary_role_Specialist Bowler`, etc.) with `drop_first=True` to avoid the **dummy variable trap** (multicollinearity).

- **HOW is it implemented?**

Using Pandas `pd.get_dummies(df, columns=['primary_role'], drop_first=True)`.

3. Supervised Learning: Continuous Regression Models

Regression models predict a continuous target value: an **Overall Performance Rating (50.0 – 95.0)** that powers fair auction purse estimation in ₹ Crores.

3.1 Random Forest Regressor — 🏆 PRODUCTION REGRESSION ENGINE

- **WHAT is it?**

An ensemble of **150 randomized decision regression trees** that each predict a player's continuous rating. The final rating is the average prediction across all 150 trees.

- **WHEN is it used?**

Powers the **Live AI Valuation Engine (Tab 4)** and the **Regression Feature Importance Leaderboard (Tab 6)**.

- **WHY do we need it?**

1. **Non-Linear Interactions:** Handles complex cricket relationships (e.g. high strike rate is great, but only when paired with decent average).
2. **Robust Against Overfitting:** Averaging 150 bootstrapped trees prevents any single outlier from skewing ratings.
3. **High Generalization:** Achieved 95.22% 5-fold cross-validation R^2 .

- **HOW does it work & performance?**

- Initialized with `RandomForestRegressor(n_estimators=150, max_depth=10, random_state=42)`.
 - **Test R^2 Score:** `0.9780` (explains 97.8% of rating variance on unseen players).
 - **5-Fold CV R^2 :** `0.9522 ± 0.0128` (highly consistent across folds).
 - **Test RMSE:** `1.2837` points | **Test MAE:** `0.9162` points.
 - **Top Factors:** Clutch Index (57.4%), Batting Impact Index (20.6%), Matches Played (4.4%).
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3.2 Polynomial Regression (Degree 2) — 🏆 TOP STATISTICAL FIT

- **WHAT is it?**

An extension of linear regression that adds squared feature terms (x_i^2) and interaction pairs ($x_i \cdot x_j$) to capture curved, non-linear relationships.

- **WHEN is it used?**

Benchmarked in Tab 6 as the high-capacity non-linear polynomial baseline.

- **WHY do we need it?**

Cricket performance follows diminishing returns or exponential growth: hitting boundaries at the death yields non-linear increases in match win probabilities.

- **HOW does it work & performance?**

- Created using `PolynomialFeatures(degree=2, include_bias=False)` piped into `Ridge(alpha=100.0)`.
 - **Test R^2 Score:** `0.9941` (highest mathematical fit).
 - **5-Fold CV R^2 :** `0.9956 ± 0.0012`.
 - **Test RMSE:** `0.6648` points | **Test MAE:** `0.3647` points.
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3.3 Linear Regression (Ordinary Least Squares – OLS)

- **WHAT is it?**

A fundamental parametric model that calculates a weighted sum of all input features to predict rating:

$$\hat{y} = \beta_0 + \sum_{j=1}^p \beta_j x_j$$

- **WHEN is it used?**

Used as the baseline benchmark in Tab 6 to verify whether simpler linear relationships can explain rating.

- **WHY do we need it?**

Direct interpretability: each coefficient β_j represents the marginal increase in player rating per standard deviation increase in that feature.

- **HOW does it work & performance?**

- Fit analytically via Scikit-Learn `LinearRegression()`.
 - **Test R^2 Score:** `0.9490`.
 - **5-Fold CV R^2 :** `0.8797 ± 0.0371`.
 - **Test RMSE:** `1.9534` points | **Test MAE:** `1.5364` points.
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4. Supervised Learning: Multi-Class Classification Models

Classification models categorize each cricketer into one of **3 discrete talent tiers**:

- **Tier 1 (Elite / Marquee):** High-impact franchise anchors and match winners.
- **Tier 2 (Core / Star):** Reliable starting XI performers and role specialists.

- **Tier 0 (Developing / Squad):** Emerging young prospects and backup squad depth.
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4.1 K-Nearest Neighbors (KNN, $k=7$) — 🏆 TOP TEST ACCURACY

- **WHAT is it?**

A non-parametric model that classifies a player by finding their **7 closest historical peers** in 29-dimensional space and taking a distance-weighted majority vote.

- **WHEN is it used?**

Evaluated in Tab 6 as the top test accuracy model for talent tier classification.

- **WHY do we need it?**

1. **Top Accuracy:** Reached **95.16% test accuracy** and **0.9501 Macro F1**.

2. **Mirrors Real Cricket Scouting:** When talent scouts evaluate a domestic prospect, they naturally ask: *"Who does this player look like in the historical record?"* (e.g. matching Andre Russell or Jasprit Bumrah).

- **HOW does it work & performance?**

- Computes Euclidean distance: $d(p, q) = \sqrt{\sum (p_i - q_i)^2}$.

- **Test Accuracy:** 95.16% | **Macro F1:** 0.9501 | **Precision:** 0.9601 | **Recall:** 0.9421 .

- **5-Fold CV Accuracy:** 90.30% ± 3.97% .

4.2 Random Forest Classifier — 🏆 PRODUCTION TIER ENGINE

- **WHAT is it?**

An ensemble of **150 randomized decision trees** voting on a player's tier.

- **WHEN is it used?**

Powers the **Live AI Valuation Engine (Tab 4)** tier classification and **Classification Feature Importance Leaderboard (Tab 6)**.

- **WHY do we need it?**

1. **High Cross-Validation Stability:** Achieved **94.75% 5-fold CV score** with minimal variance ($\pm 2.25\%$).

2. **Gini Impurity Feature Importance:** Explicitly reveals which measurable factors drive elite tier categorization.

- **HOW does it work & performance?**

- Initialized with `RandomForestClassifier(n_estimators=150, max_depth=10, random_state=42)` .

- **Test Accuracy:** 94.35% | **5-Fold CV Score:** 94.75% | **Macro F1:** 0.9459 .

- **Top Factors:** Clutch Index (12.8%), Batting Impact Index (11.6%), Player of Match Awards (7.8%), Matches Played (6.8%).

4.3 Decision Tree Classifier (CART)

- **WHAT is it?**

A hierarchical flowchart of logical "IF-THEN" rules splitting data on Gini impurity.

- **WHEN is it used?**

Benchmarked in Tab 6 for maximum human interpretability.

- **WHY do we need it?**

Enables coaching staff and non-technical executives to audit exact decision thresholds without black-box opacity.

- **HOW does it work & performance?**

- Fit with `DecisionTreeClassifier(max_depth=5, random_state=42)` .

- **Test Accuracy:** 87.90% | **5-Fold CV Score:** 89.90% | **Macro F1:** 0.8812 .
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4.4 Logistic Regression (Multinomial / Softmax)

- **WHAT is it?**

A linear classification model that maps feature combinations into calibrated class probabilities that sum to 100%.

- **WHEN is it used?**

Provides the probability distribution displayed in the Live AI Valuation Engine (Tab 4).

- **WHY do we need it?**

Franchise directors managing bidding caps need risk percentages (e.g. 85\% Elite vs. 15\% Core) rather than rigid binary labels.

- **HOW does it work & performance?**

- Trained via `LogisticRegression(max_iter=1000, random_state=42)` .

- **Test Accuracy:** 87.90% | **5-Fold CV Score:** 92.53% | **Macro F1:** 0.8853 .
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5. Unsupervised Learning: Clustering (Tactical Archetypes)

Unlike classification, clustering does not use human labels. It automatically groups athletes by multidimensional skill fingerprints.

5.1 K-Means Clustering ($k=5$)

- **WHAT is it?**

An iterative clustering algorithm that partitions 619 cricketers into **5 tactical archetypes** by minimizing within-cluster sum of squares (inertia):

$$\mathcal{J} = \sum_{i=1}^k \sum_{x \in S_i} \|x - \mu_i\|^2$$

- **WHEN is it used?**

Powers **Tab 5 (Tactical Archetypes & Roster Balance)**.

- **WHY do we need it?**

Nominal labels ("Batter" or "Bowler") fail modern franchise recruitment:

- Virat Kohli and Andre Russell are both "Batters", but Kohli anchors innings while Russell finishes in death overs.
- Jasprit Bumrah and Yuzvendra Chahal are both "Bowlers", but Bumrah bowls 145 km/h yorkers while Chahal controls middle overs with leg-spin.
- K-Means automatically discovered these **5 tactical archetypes**:

1. **Cluster 0: Top-Order Anchor & Accumulator** (high average, builds partnerships).
2. **Cluster 1: Powerplay & Middle-Overs Pace Specialist** (wicket-taking seam bowling).
3. **Cluster 2: Death-Overs Finisher & Power Hitter** (strike rate >150, high boundary %).
4. **Cluster 3: Defensive Middle-Overs Economy Spinner** (low economy rate, high dot ball %).
5. **Cluster 4: Elite Dual-Threat All-Rounder** (high contributions with both bat and ball).

- **HOW does it work?**

- Validated using the Elbow Method and Silhouette Analysis ($s = 0.2676$).
 - Serialized as `models/kmeans_model.joblib` .
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6. Dimensionality Reduction: Visualizing High-Dimensional Data

6.1 Principal Component Analysis (PCA, 2D Latent Space)

- **WHAT is it?**

An orthogonal linear transformation that projects 29 performance metrics onto 2 uncorrelated principal components (PC_1, PC_2).

- **WHEN is it used?**

Powers the interactive 2D Tactical Scatter Plot in **Tab 5**.

- **WHY do we need it?**

Human decision-makers cannot comprehend a 29-dimensional space. PCA compresses the data into an intuitive 2D tactical map where coaching staff can instantly see tactical proximity between any two players.

- **HOW does it work & performance?**

- Derived via Singular Value Decomposition (SVD) of the covariance matrix.
- **PC_1 (X-axis, 38.16% variance)**: Measures career volume, longevity, and overall match contribution.
- **PC_2 (Y-axis, 22.63% variance)**: Captures skill orientation (bowling-dominant at top, batting-dominant at bottom, dual-threat in middle).
- **Total Variance Explained**: 60.79% across the first two components.
- Serialized as `models/pca_model.joblib`.

7. Model Evaluation & Validation Rigor

To prevent data leakage and guarantee that models generalize to future tournaments:

1. **80/20 Stratified Split**: 495 training players (80%) and 124 held-out test players (20%) with identical tier proportions.

2. **5-Fold Stratified Cross-Validation**: Every model is evaluated across 5 distinct validation slices to prove score stability.

3. **Comprehensive Metric Suite**:

- **Regression**: R^2 (variance explained), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE).
 - **Classification**: Accuracy, Precision, Recall, Macro F1-Score, and Full Confusion Matrices.
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8. Master Model Leaderboard

Model	Task	Test Metric	5-Fold Cross-Validation	Key Strength / Role in System
Polynomial Reg (Deg 2)	Continuous Rating (50-95)	$R^2 = 0.9941$ (RMSE: 0.6648)	0.9956 ± 0.0012	Mathematical precision modeling curved feature synergies.
Random Forest Regressor	Continuous Rating (50-95)	$R^2 = 0.9780$ (RMSE: 1.2837)	0.9522 ± 0.0128	Production rating engine & continuous feature rankings.
Linear Regression (OLS)	Continuous Rating (50-95)	$R^2 = 0.9490$ (RMSE: 1.9534)	0.8797 ± 0.0371	Parametric baseline measuring direct linear marginal impacts.
K-Nearest Neighbors (KNN)	Talent Tier Classification	95.16% Acc (F1: 0.9501)	0.9030 ± 0.0397	Top test accuracy; identifies nearest historical player comparables.
Random Forest Classifier	Talent Tier Classification	94.35% Acc (F1: 0.9459)	0.9475 ± 0.0225	Top CV stability; production tier classification & Gini rankings.
Logistic Regression	Talent Tier Classification	87.90% Acc (F1: 0.8853)	0.9253 ± 0.0310	Outputs calibrated auction risk probabilities (0-100%).
Decision Tree (CART)	Talent Tier Classification	87.90% Acc (F1: 0.8812)	0.8990 ± 0.0293	Transparent "IF-THEN" rule tree for coaching audits.
K-Means Clustering	Tactical Playing Styles	$k = 5$ Archetypes ($s = 0.2676$)	Elbow / Silhouette	Groups players by true tactical capability beyond nominal roles.
PCA (2 Components)	2D Visualization	60.79% Variance	SVD Eigenvalues	Projects 29-dimensional performance into an interpretable 2D map.

Key Takeaway for Viva & Presentation Defense

Question: "Why did you use both regression and classification models in this system?"

Answer:

*"In a sports franchise front office, decision-makers require two distinct levels of precision:

1. **Classification (KNN & Random Forest Classifier)** solves squad composition by grouping players into 3 broad talent tiers (Elite, Core, Developing) for roster depth and retention limits.
2. **Regression (Random Forest Regressor & Polynomial Regression)** solves financial valuation by computing a precise, continuous FIFA-style rating (50.0 - 95.0) that directly calculates a fair auction purse in ₹ Crores, ensuring the franchise never overpays in bidding wars."